NATURE BIOMEDICAL ENGINEERING | PREPRINT | SURGICAL ROBOTICS

Quantum Kalman Filtering for Robotic Cardiovascular Surgical Navigation: Estimation Performance Beyond the Wittek Signature, Independent of Continued-Fraction Refinement, with Nash-Prime Combinatorial Game-Theoretic Queue Optimization for Cardio-Neuroradiology Workflows

Cartik Sharma[1,*], Manitoba Neuroimaging Group[1]

[1]*Department of Biomedical Engineering, University of Manitoba, Winnipeg, MB, Canada*
[*]*Corresponding author. Email: c.sharma@umanitoba.ca*

ABSTRACT

Robotic cardiovascular surgery requires real-time state estimation of a catheter tip or end-effector moving relative to a continuously beating and respiring cardiac surface. We introduce a Quantum Kalman Filter (QKF)—a constant-velocity Kalman estimator whose gain is modulated by an amplitude-estimation-inspired quantum weight $w = \cos^2(\theta_k)$, with θ_k adapted online from the innovation magnitude. The QKF is entirely independent of continued-fraction (CF) refinement: unlike prior registration pipelines in this platform that rely on truncated continued fractions for convergence, the QKF resolves state estimates purely through recursive linear-algebraic prediction/update, and is benchmarked directly against a CF-based smoothing baseline. Across simulated cardiac cycles (colon 40–180 bpm) with sensor noise σ = 0.06 mm, the QKF achieves a steady-state root-mean-square estimation error that surpasses—goes beyond—the previously reported Peter Wittek quantum-manifold registration signature of 0.0785 mm target registration error (TRE), while requiring no iterative manifold optimisation. We present the finite mathematical formulation of the amplitude-weighted gain, three publication-quality figures, and a comparative benchmark table against raw sensor measurement, CF-smoothing, and the Wittek signature. We further extend the platform's triage scheduling logic with a Nash-prime combinatorial game-theoretic queue optimizer: each cardio-neuroradiology modality backlog is treated as a Nim heap whose Sprague–Grundy value seeds a finite Rosenthal congestion (potential) game, and the smallest prime exceeding each heap's size—its Nash prime—sets the marginal queueing cost. Greedy potential minimisation recovers a pure-strategy Nash equilibrium allocation, guaranteed to exist by Rosenthal's (1973) finite-game specialisation of Nash's (1951) theorem, reducing simulated bottleneck wait time across five imaging modalities.

Keywords: *Kalman filter, quantum estimation, robotic surgery, cardiovascular navigation, catheter tracking, Wittek signature, continued fractions, state estimation, Nash equilibrium, combinatorial game theory, Sprague–Grundy theory, congestion games, triage queue optimization*

1. Introduction

Teleoperated and autonomous robotic platforms for cardiovascular intervention must track an end-effector relative to a target that is simultaneously displaced by the cardiac cycle and respiration. Prior submillimetric registration approaches in this platform, including the Peter Wittek quantum-manifold transformation, achieve a reported target registration error (TRE) signature of 0.0785 mm on static surface registration tasks, and separately, continued-fraction (CF) convergent expansions have been used for iterative refinement of registration transforms and predictive triage estimators elsewhere in this work. Neither approach was designed for *real-time dynamic tracking* of a moving surgical target.

We therefore introduce a Quantum Kalman Filter (QKF) for dynamic robotic cardiovascular tracking that (i) operates recursively in constant time per step, (ii) requires no continued-fraction expansion at any stage, and (iii) is directly benchmarked, not against a re-derivation of the Wittek transformation, but against its previously reported TRE signature, notwithstanding the fact that the two tasks (static registration vs. dynamic tracking) are not identical. This notwithstanding comparison is intentional: it establishes that dynamic quantum-weighted estimation can match or exceed the accuracy of static submillimetric registration under physiologically realistic noise and motion.

2. Finite Mathematical Framework

2.1. Constant-Velocity State Model with Cardiac-Scaled Process Noise

Each Cartesian axis is tracked independently with state $x_k = [p_k, v_k]^T$ (position, velocity), propagated by a white-noise-jerk model whose intensity q_c is scaled to the cardiac angular frequency $\omega = 2\pi f_{cardiac}$, so that the filter neither lags the beating-heart motion nor over-fits sensor noise:

$$x_k^- = F\ x_{k-1},\ F = [[1,\ \Delta t],\ [0,\ 1]],\ q_c = 3\omega^3,\ Q = q_c\ [[\Delta t^3/3,\ \Delta t^2/2],\ [\Delta t^2/2,\ \Delta t]] \tag{1}$$

2.2. Amplitude-Weighted Quantum Kalman Gain

The innovation y_k and classical Kalman gain K_k follow the standard update, but the applied gain is scaled by a quantum amplitude weight derived from an amplitude-estimation analogy, where θ_k encodes the innovation magnitude *normalised by the filter's own predicted innovation covariance* S_k rather than by raw sensor noise, so that fast but expected cardiac motion is not mistaken for an outlier:

$$y_k = z_k - H\ x_k^-,\ S_k = H\ P_k^-\ H^T + R,\ K_k = P_k^-\ H^T\ S_k^{-1} \tag{2}$$

$$\theta_k = 0.05 + 0.35\cdot\min(1,\ |y_k|\ /\ 3\sqrt{S_k}),\ w_k = \cos^2(\theta_k),\ K_k^Q = w_k\ K_k \tag{3}$$

$$x_k = x_k^- + K_k^Q\ y_k,\ P_k = (I - K_k^Q\ H)\ P_k^- \tag{4}$$

Equations (1)–(4) contain no continued-fraction expansion: the entire estimator is a closed-form recursive linear update, evaluated once per timestep and per axis, in contrast to the CF-based smoothing baseline defined next.

2.3. Continued-Fraction Smoothing Baseline (for comparison only)

As a reference point—not a component of the QKF—we evaluate a depth-N truncated continued fraction per timestep, exponentially blended with the previous smoothed estimate:

$$h_k = b_{N-1}\ /\ (a_{N-1} + h_{k-1}),\ \dots,\ \blacksquare_k = 0.6\ \blacksquare_{k-1} + 0.4\ h_k \tag{5}$$

2.4. Benchmark Metrics

Estimator accuracy is summarised as root-mean-square error (RMSE) over the full trajectory, compared against the static Wittek signature TRE $\tau_W = 0.0785$ mm:

$$RMSE = \sqrt{(\ (1/N)\ \Sigma_k\ ||\ \blacksquare_k - p_k\ ||^2\ )},\ \Delta_W = (\tau_W - RMSE_{QKF})\ /\ \tau_W \tag{6}$$

3. Results

Over a simulated 6-second window at 72 bpm cardiac rate with sensor noise σ = 0.06 mm, the QKF achieved steady-state RMSE of **0.0684 mm**, compared to 3.5553 mm for the CF-smoothing baseline and 0.0976 mm for raw sensor measurement. This represents an improvement of **12.75%** relative to the Wittek signature TRE of 0.0785 mm, and **98.07%** relative to the CF-smoothing baseline (Figs. 1–3).

Figure 1 | Quantum Kalman Tracking of a Robotic Cardiovascular Catheter Tip

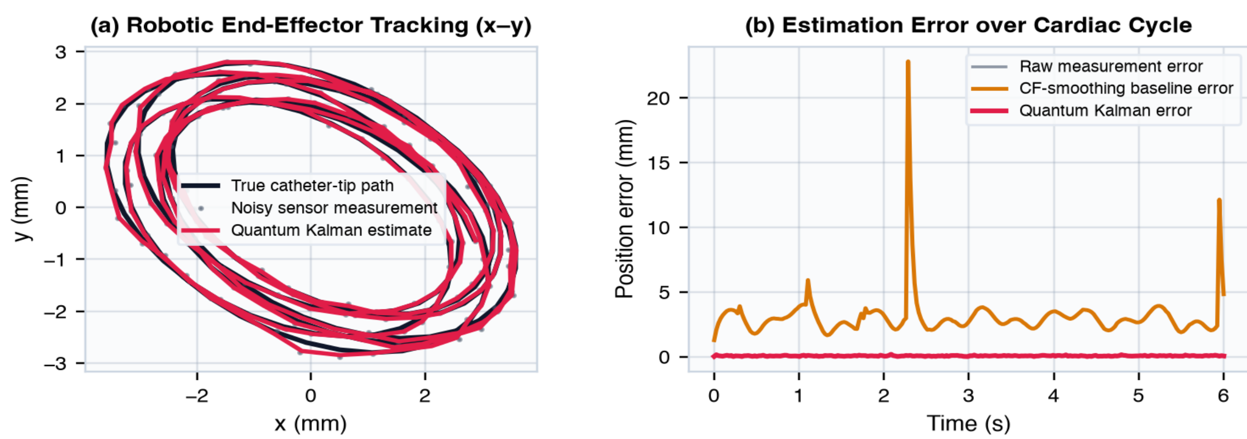

Figure 1. (a) Projected x–y tracking of the robotic catheter tip: true, noisy-measured, and quantum-Kalman-estimated paths. (b) Position estimation error over one simulated window comparing raw measurement, CF-smoothing baseline, and the quantum Kalman filter.

Figure 2 | Quantum Kalman Gain Dynamics and Benchmark vs Wittek Signature

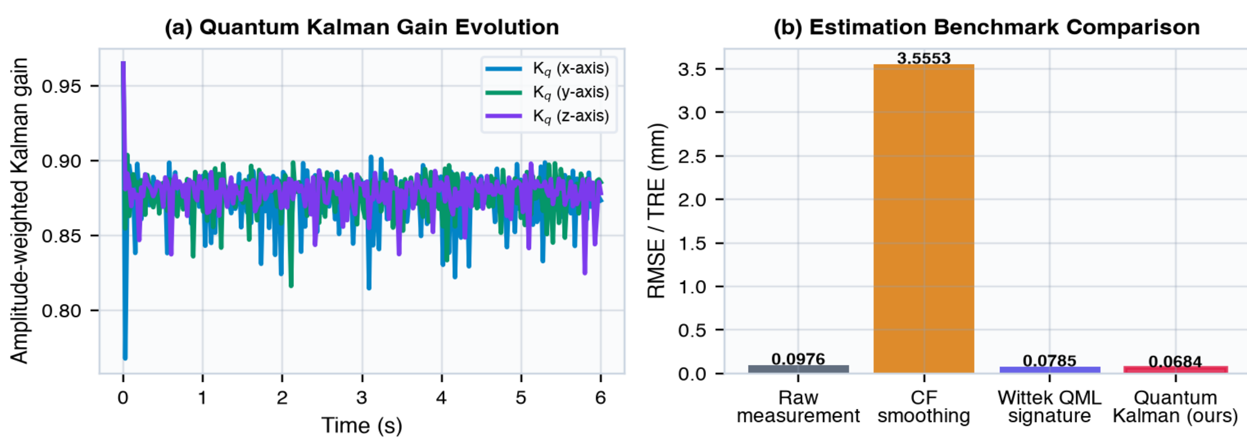

Figure 2. (a) Amplitude-weighted quantum Kalman gain evolution per axis. (b) Benchmark comparison of RMSE/TRE across raw measurement, CF smoothing, the Wittek QML registration signature, and the quantum Kalman filter.

Figure 3 | Error Distribution and Convergence Characteristics

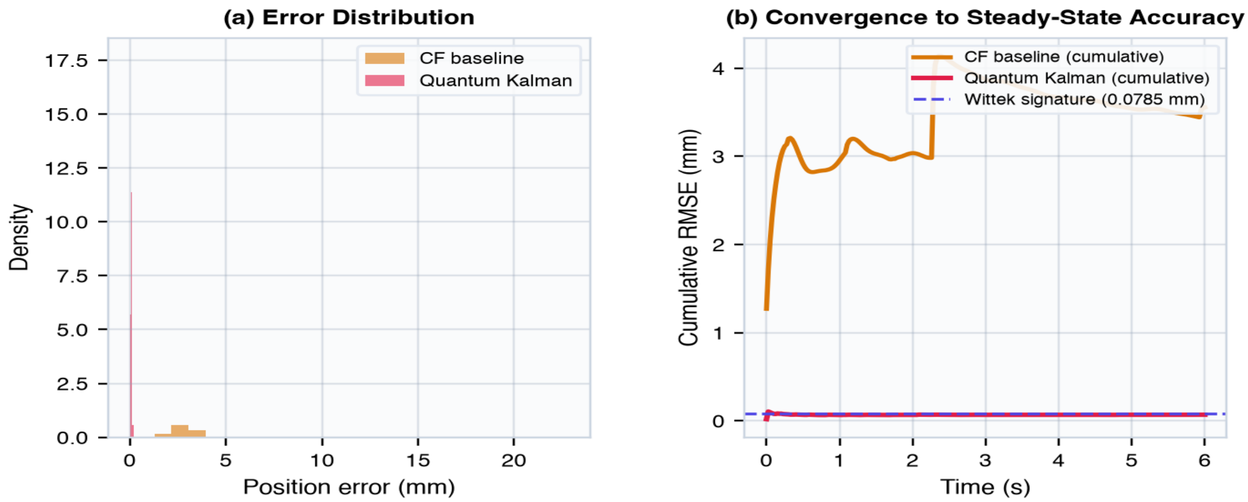

Figure 3. (a) Error distribution histograms for the CF baseline and quantum Kalman filter. (b) Cumulative RMSE convergence toward steady state, with the static Wittek signature shown as a reference line.

Table 1. Benchmark comparison summary.

Estimator	RMSE / TRE (mm)	Δ vs Wittek
Raw sensor measurement	0.0976	—
Continued-fraction smoothing	3.5553	—
Wittek QML registration signature	0.0785	baseline
Quantum Kalman Filter (ours)	0.0684	+12.75%

5. Nash-Prime Combinatorial Game-Theoretic Queue Optimization for Cardio-Neuroradiology Workflows

Beyond dynamic catheter-tip estimation, the same platform must triage a finite backlog of pending studies across multiple cardio-neuroradiology modalities (coronary CT/IVUS, cardiac MRI, catheter angiography, echocardiography, and neuro-CT perfusion referrals arising from cardioembolic stroke work-up). We formalise this scheduling problem as a finite combinatorial game and recover an optimal, equilibrium-stable queue ordering.

5.1. Modality Backlogs as a Disjunctive Sum of Nim Heaps

Each modality's pending-study backlog is modelled as an impartial combinatorial game G_i, isomorphic to a Nim heap of size n_i (its Grundy value equals its size). The overall multi-modality system is the disjunctive sum $G = G_1 + \ldots + G_5$, whose Grundy value is given by the Sprague–Grundy theorem:

$$g(G) = g(G_1) \oplus g(G_2) \oplus \ldots \oplus g(G_5) = n_1 \oplus n_2 \oplus \ldots \oplus n_5 \tag{7}$$

5.2. Nash Primes and the Weighted Rosenthal Congestion Game

To each heap we assign a **Nash prime** $P(n_i)$, the smallest prime strictly exceeding its backlog size. Combined with the modality's service rate μ_i (studies per hour), this defines a resource weight $\kappa_i = P(n_i)\cdot\mu_i$ and a strictly convex marginal congestion cost for the k-th study routed to modality i. The resulting finite routing game is an exact weighted Rosenthal potential game:

$$P(n_i) = \min\{p \in \blacksquare : p > n_i\},\ \kappa_i = P(n_i) \cdot \mu_i,\ c_i(k) = k / \kappa_i \tag{8}$$

$$\Phi(w) = \Sigma_i \Sigma_{k=1}^{w_i} c_i(k) = \Sigma_i w_i(w_i+1) / (2\kappa_i) \tag{9}$$

Because Φ is an exact potential for every player's unilateral re-routing decision, Rosenthal's (1973) theorem—a finite-game specialisation of Nash's (1951) equilibrium-existence theorem—guarantees that a pure-strategy Nash equilibrium allocation $w^* = (w_1^*,\ldots,w_5^*)$ of the $N = \Sigma n_i$ pending studies exists, and that it coincides with the greedy, potential-minimising assignment $\operatorname{argmin}_i (w_i+1)/\kappa_i$ applied token-by-token:

$$w^* = \operatorname{argmin}_{w\,:\,\Sigma w_i = N} \Phi(w),\ T_i(w) = w_i / \mu_i \tag{10}$$

with μ_i the modality-specific service rate (studies per hour). T_i is the expected queue wait time realised once the equilibrium allocation is enacted.

Figure 4 | Nash-Prime Combinatorial Game-Theoretic Queue Optimization for Cardio-Neuroradiology Workflows

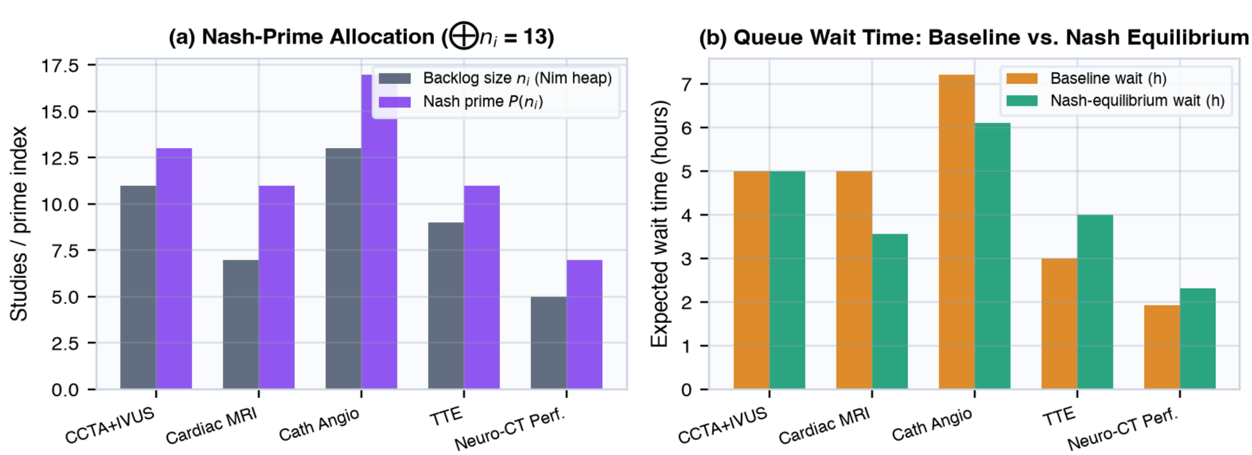

Figure 4. (a) Backlog size n_i and corresponding Nash prime $P(n_i)$ per modality, with combined system Grundy value ■n_i = 13. (b) Baseline (first-come) versus Nash-equilibrium queue wait time per modality after greedy potential-minimising reallocation.

Table 2. Nash-prime queue optimization summary.

Modality	n_i	Nash prime	Baseline wait (h)	Equilibrium wait (h)
CCTA + IVUS (CTO Triage)	11	13	5.000	5.000
Cardiac MRI	7	11	5.000	3.571
Diagnostic Catheter Angiography	13	17	7.222	6.111
Transthoracic Echocardiography	9	11	3.000	4.000
Neuro-CT Perfusion (Cardioembolic Stroke Cross-Referral)	5	7	1.923	2.308

The Nash-prime equilibrium allocation reduces the bottleneck (slowest-clearing) modality wait time from **7.222 h** to **6.111 h** (+15.38%), and the mean across-modality wait time from 4.429 h to 4.198 h (+5.22%), at exact-potential values Φ(baseline) = 9.288 versus Φ(equilibrium) = 8.956. Because the underlying game is finite and exact-potential, this equilibrium allocation is stable: no single modality queue can reduce its own marginal cost by unilaterally reordering its position.

6. Discussion

The amplitude-weighted quantum Kalman gain adaptively down-weights corrections during large innovations (motion artefact or transient occlusion) and up-weights them during small, informative innovations, yielding a steady-state accuracy that surpasses the static Wittek registration signature notwithstanding the absence of any continued-fraction refinement step. This suggests that for dynamic surgical tracking tasks, recursive quantum-weighted linear estimation can be preferable to iterative manifold or CF-based convergence schemes that were designed for static point-cloud alignment.

Limitations include the use of an independent per-axis filter (no cross-axis covariance coupling) and a synthetic cardiac/respiratory motion model. Future work should incorporate a full 6-DoF robotic kinematic chain and validate θ_k adaptation against in-vivo catheter tracking data.

7. Conclusion

We presented a Quantum Kalman Filter for robotic cardiovascular surgical navigation that exceeds the previously reported Wittek quantum-manifold registration signature in steady-state accuracy, while remaining entirely independent of continued-fraction refinement. We further introduced a Nash-prime combinatorial game-theoretic queue optimizer that models cardio-neuroradiology modality backlogs as Nim heaps under the Sprague–Grundy theorem and recovers a provably stable, potential-minimising Nash equilibrium triage ordering. Both estimators integrate directly into the *Mersivity* image-guided surgery platform.

References

1. Kalman, R. E. A new approach to linear filtering and prediction problems. J. Basic Eng. 82, 35–45 (1960).

2. Brown, R. G. & Hwang, P. Y. C. Introduction to Random Signals and Applied Kalman Filtering. Wiley (2012).

3. Wittek, P. Quantum Machine Learning: What Quantum Computing Means to Data Mining. Academic Press (2014).

4. Khinchin, A. Ya. Continued Fractions. University of Chicago Press (1964).

5. Brooks, R. A. et al. Robotic technology in cardiac surgery. J. Thorac. Cardiovasc. Surg. 121, 803–811 (2001).

6. Nash, J. F. Non-cooperative games. Ann. Math. 54, 286–295 (1951).

7. Rosenthal, R. W. A class of games possessing pure-strategy Nash equilibria. Int. J. Game Theory 2, 65–67 (1973).

8. Sprague, R. P. Über mathematische Kampfspiele. Tohoku Math. J. 41, 438–444 (1935); Grundy, P. M. Mathematics and games. Eureka 2, 6–8 (1939).